

Interestingness Measure: Lift

- Measure of dependent/correlated events: **lift**

$$\text{lift}(B, C) = \frac{c(B \rightarrow C)}{s(C)} = \frac{s(B \cup C)}{s(B) \times s(C)}$$

- Lift(B, C) may tell how B and C are correlated

- Lift(B, C) = 1: B and C are independent
- > 1: positively correlated
- < 1: negatively correlated

- For our example,

$$\text{lift}(B, C) = \frac{400/1000}{600/1000 \times 750/1000} = 0.89$$

$$\text{lift}(B, \neg C) = \frac{200/1000}{600/1000 \times 250/1000} = 1.33$$

- Thus, B and C are negatively correlated since $\text{lift}(B, C) < 1$;
 - B and $\neg C$ are positively correlated since $\text{lift}(B, \neg C) > 1$

Lift is more telling than s & c

	B	$\neg B$	Σ_{row}
C	400	350	750
$\neg C$	200	50	250
$\Sigma_{\text{col.}}$	600	400	1000